Industry ML Case Study – ML-021

Enterprise Supply Chain Demand Forecasting & Inventory Optimization Platform

Duration

2 Weeks

Industry Background

Global retailers, manufacturers, and logistics companies manage millions of products across hundreds of warehouses. Incorrect demand forecasting leads to stockouts, overstocking, increased storage costs, lost sales, and inefficient supply chain operations.

Organizations require an AI-powered forecasting platform capable of predicting product demand, optimizing inventory levels, and recommending replenishment strategies.

Business Problem

A multinational retail company operates:

- 250 Warehouses
- 4 Million Products
- 45 Countries
- 30 Million Customers
- 500 Million Orders per Year

Current challenges include:

- Frequent stock shortages
- Overstocked warehouses

- Seasonal demand fluctuations
- Supplier delays
- Regional demand differences
- High inventory holding costs
- Poor warehouse utilization

The company requires an intelligent forecasting and inventory optimization platform.

Project Objective

Develop an **Enterprise Supply Chain Demand Forecasting & Inventory Optimization Platform** capable of forecasting product demand, optimizing inventory distribution, estimating reorder points, detecting abnormal demand spikes, and supporting procurement planning.

Core Modules

Enterprise Data Pipeline

Collect:

- Historical Sales
- Warehouse Inventory
- Supplier Performance
- Product Master Data
- Promotions
- Holidays
- Weather Data
- Regional Events
- Delivery Times

Feature Engineering

Generate:

- Sales Velocity

- Seasonality Index
- Promotion Impact
- Lead Time
- Supplier Reliability Score
- Product Lifecycle Stage
- Demand Volatility
- Regional Purchasing Patterns
- Holiday Effects

Demand Forecasting Engine

Predict:

- Daily Demand
- Weekly Demand
- Monthly Demand
- Seasonal Peaks
- Regional Demand
- Product Category Demand

Inventory Optimization Engine

Recommend:

- Reorder Point
- Safety Stock
- Economic Order Quantity (EOQ)
- Warehouse Allocation
- Inter-Warehouse Transfers
- Procurement Quantity

Supply Chain Risk Engine

Detect:

- Supplier Delay Risk
- Stockout Risk

- Overstock Risk
- Slow Moving Inventory
- Dead Inventory
- Demand Anomalies

Scenario Simulation

Simulate:

- Supplier Failure
- Price Increase
- Holiday Sales
- New Product Launch
- Transportation Delay
- Demand Surge

Predict operational impact before execution.

Executive Dashboard

Display:

- Forecast Accuracy
- Inventory Health
- Warehouse Utilization
- Fill Rate
- Stockout Probability
- Procurement Recommendations
- Supply Chain KPIs

Alert Center

Generate alerts for:

- Low Inventory
- Overstock
- Demand Spike
- Supplier Delay

- Warehouse Capacity Risk
- Forecast Drift

Machine Learning Components

Implement:

- XGBoost
- LightGBM
- Prophet
- LSTM (Optional)
- CatBoost (Optional)
- Time Series Cross Validation
- Feature Selection
- Bayesian Hyperparameter Optimization

MLOps Components

Include:

- MLflow
- Model Registry
- Data Drift Detection
- Automated Retraining
- Model Monitoring
- Experiment Tracking

Technical Requirements

Backend

- Python
- FastAPI
- PostgreSQL

- Redis

ML Stack

- Pandas
- NumPy
- Scikit-learn
- XGBoost
- LightGBM
- Prophet
- MLflow
- SHAP

Non-Functional Requirements

Support:

- 100 Million Forecast Requests
- Real-Time Prediction APIs
- High Availability
- Horizontal Scaling
- Explainable Forecasts
- Secure API Access

Deliverables

- Complete Source Code
- Forecasting Pipeline
- Feature Engineering Pipeline
- Inventory Optimization Engine
- FastAPI Services
- Dashboard
- MLOps Pipeline
- Model Evaluation Report
- Architecture Diagram
- Deployment Guide

- README
- Technical Presentation

Bonus Challenges

Implement one or more:

- Reinforcement Learning for Inventory Policies
- Multi-Echelon Inventory Optimization
- Graph Neural Networks for Supply Chain Networks
- Carbon Footprint Optimization
- Digital Twin Warehouse Simulation
- Supplier Recommendation Engine

Evaluation Criteria

Criteria	Weight
Feature Engineering	20%
Forecast Accuracy	20%
Inventory Optimization Logic	20%
MLOps Implementation	15%
Explainable AI	10%

Documentation	10%
Presentation	5%

Expected Learning Outcomes

By completing this project, interns should demonstrate the ability to:

- Build enterprise demand forecasting systems.
- Engineer complex time-series features from supply chain data.
- Optimize inventory using predictive analytics.
- Deploy scalable forecasting APIs with MLOps best practices.
- Design AI systems that support enterprise supply chain decision-making.